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Research on the perception capability and characteristics based on the full-attitude indirect sensing model of forging force

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ABSTRACT

In free forging production, the time-varying force exerted by the forging on the forging manipulator is highly dynamic, with severe impacts and harsh operating conditions, making direct measurement challenging. To address this issue, this study proposes a full-attitude indirect sensing model of forging force based on the hinge-point forces of the clamping mechanism. The model further takes into account the effects of the clamping posture, the equivalent diameter of the forging, and other state variables. The accuracy of the full-attitude hinge-point force model has been validated using a prototype, with the worst-case error being 9.3%, enabling high-precision description of hinge-point forces under full attitude. Based on the prototype and the model, the sensing capabilities and patterns of hinge-point forces on the forging under different operating conditions were analyzed. It was found that hinge-points O, O', C, and C' exhibit stronger sensing capabilities. This study identified the appropriate hinge-points and force components for sensing different conditions and information, providing a theoretical foundation for the forging force identification and buffering in forging manipulators.

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Forging manipulator;
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forging force; indirect
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force model

1. Introduction

The forging industry plays a pivotal role in the national economy. Forging manipulators are important auxiliary equipment in free forging workshops, working in conjunction with presses. They are mainly used for clamping forgings and flip operations, which are of great significance for improving the quality of forged products and increasing forging production efficiency (Ge and Gao 2012). During the forging process, forgings exert forces on the clamping mechanism of the forging manipulator. In this article, these forces are defined as forging forces. From a systematic perspective, forging forces manifest in the form of internal forces generated through three main pathways:

1. The resistance stemming from the deformation of the forging itself. During forging, the part of the forging placed between the manipulator and the press anvil is subjected to the combined action of the downward pressure of the press and the constraint of the manipulator,

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and the deformation or deformation trend of the forging exerts a force (or impact force) on the manipulator (Han et al. 2015).

2. When the manipulator clamps the forging and experiences impacts against the press anvil or the ground, collisions between the forging and related equipment result. Additionally, when two manipulators clamp the forging and produce coupling forces due to asynchronous movement, these forces are transmitted through the forging to the clamping mechanism of each manipulator (Zhai et al. 2022).
3. In the clamping process of the forging, gravity additionally exerts force on the manipulator via the clamping device. Notably, the first two types of forging forces present more significant risks to the free forging system. Light impacts may affect the deformation pattern of the forgings, while severe impacts can lead to equipment instability and destructive accidents.

The clamping mechanism of the forging manipulator is shown in Fig. 1. It is evident that, during forging, the constraining effect of the clamping mechanism directs the axial forging force onto the forging at the jaw, while the vertical load applied by the hydraulic press acts on the forging at the jaw. This can mildly influence the deformation pattern of the forging, and, in severe cases, it can lead to equipment instability and destructive accidents. The influence of forging force on the clamping mechanism must be taken seriously.

When confronted with the impact and potential harm resulting from forging forces, there are two main methods of mitigation: one is the passive cushioning approach, which absorbs the forging forces into energy storage devices characterized by lower stiffness. For instance, Li et al. (2016) studied the relationship between the pressure, volume, and buffering capacity of energy storage devices. The other approach is the active adaptive method, as analyzed by Li et al. (2016) who investigated the relationship between hydraulic cylinder displacement and variations in forging force, and they also employed displacement control strategies for actively adapting to forging forces. Zhang, Yao, and Du (2013) proposed an impedance model and formulated corresponding flexibility strategies to reduce the mutual reaction loads of forging manipulators. Furthermore, Hou et al. (2017) utilized buffer cylinder oil pressure as a feedback signal and designed a closed-loop control system to relieve the impact force during the forging process. The drawbacks of the aforementioned methods can be outlined as follows:

1. The intricate working conditions associated with free forging pose a challenge that is exacerbated by the constrained energy storage capacity of passive cushioning methods such as the accumulator, thus hindering comprehensive coverage across all working condition settings.
2. Despite the employment of active adaptive methods in which displacement control strategies aim to mitigate the intensity of forging forces, the prediction of deformation in complex-shaped forgings remains challenging. This challenge leads to significant uncertainty, reduced accuracy, and diminished feasibility (Wang et al. 2010).

In recent years, driven by economic growth and significant advancements in science and technology, various industries have heightened their requirements for the precision of free forgings. Especially with the emergence of some new materials, there is a growing demand for attributes such as high resistance to deformation, high precision, and a narrow forging temperature range (Papenberg et al. 2020; Hu, Yi, and Huang 2015). This highlights the shortcomings of the aforementioned cushioning methods which have become increasingly apparent. Therefore, the impact and pattern of forging forces on the clamping mechanism, emphasizing the necessity for targeted adjustments and adaptive measures based on this understanding, have become increasingly urgent (Wang et al. 2024).

The methods for elucidating forging forces include direct detection and indirect perception. Directly detecting forging forces in free forging has perennially posed a challenge in the industry,

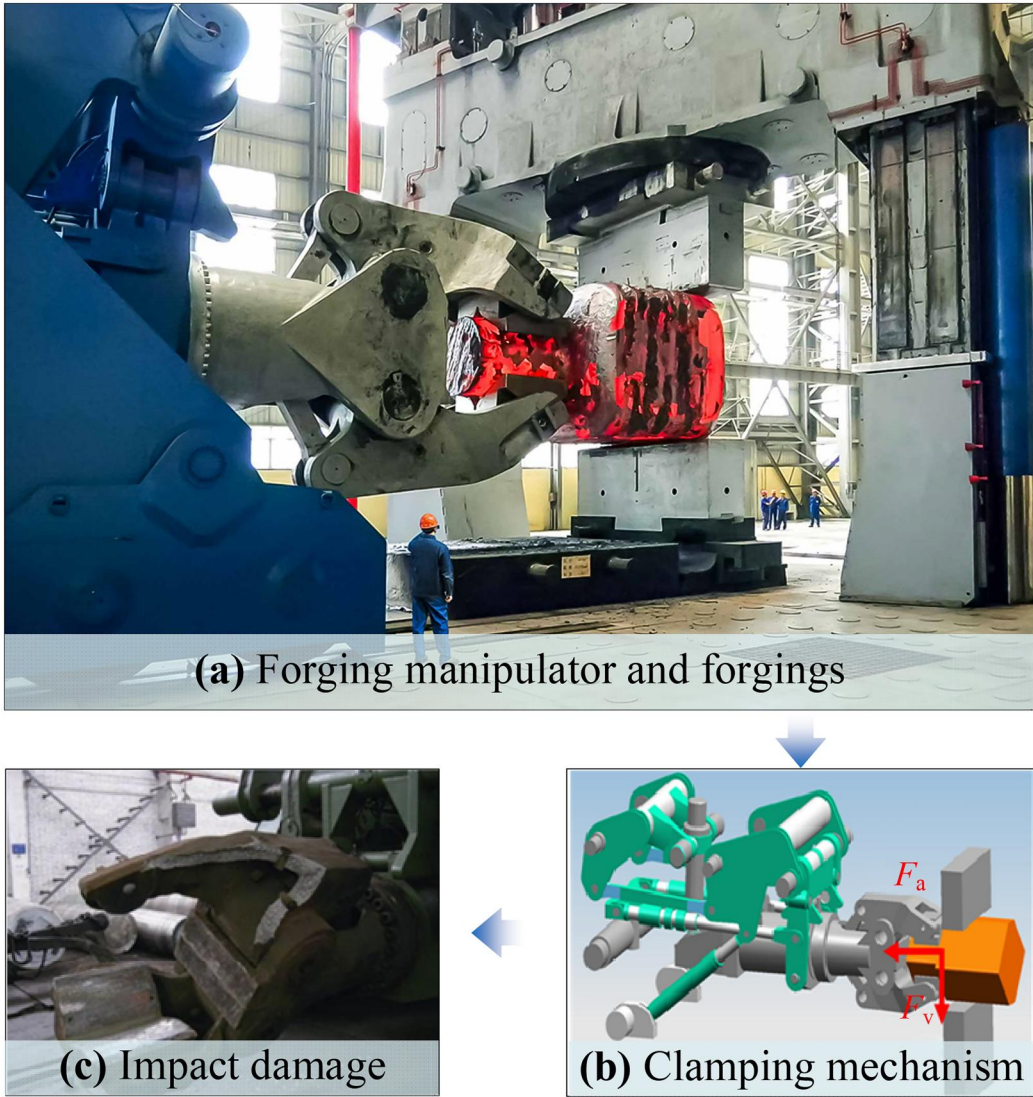


Figure 1. The clamping mechanism of the forging manipulator in forging conditions.

and examining it from various technical angles is exceedingly complex. First, due to the harsh working conditions of free forging, force detection of forgings needs to cope with high temperatures, heavy loads, and impact influences. Installing large-scale multi-dimensional force sensors on the manipulator faces safety hazards caused by reducing equipment stiffness, making direct measurement schemes impractical. In addition, the widespread introduction of the digital twin concept enables simulation and real-time prediction and control (Tang et al. 2023) and seems to offer promising prospects. However, this method also relies on data collection and perception, and, currently, real-time and accuracy capabilities are completely unattainable, as it is still confined to the offline simulation stage. In comparison, indirect sensing of forces by placing sensors in locations different from where the force of interest is located is an effective means, contrasting with direct detection.

Indirect force perception methods have been previously applied and are more mature in fields, such as medical care and precision manufacturing. Hu et al. (2019) proposed a method using

dynamic models to infer force information at the end effectors of steerable catheters/manipulators, thus enhancing the safety of operations. Cheng et al. (2013) established dynamic models providing information on the shape of endoscopes, the direction of the colon, and force, which improved treatment efficiency. Xiao et al. (2015) measured the charge signals of a knife handle and inferred cutting force through decoupling algorithms, indirectly achieving calibration and debugging of tools. This study draws on the idea of inferring the force at the end of the mechanism through detection point information. However, in terms of working conditions, such as the forging manipulator which needs to withstand impacts and overloads, and in terms of working modes, such as the rotational and postural changes of the forging manipulator, there are many differences. Currently, the concept of indirect force perception has been applied to the field of engineering equipment with working conditions similar to those of this study. The methods can be categorized into two main groups: model-based and data-driven approaches (Moussa et al. 2019).

Model-based indirect force perception methods entail the utilization of input and output signals to construct mathematical models and analyze characteristics, thus identifying mechanical trends. For instance, Yuan et al. (2019) based on an experimentally verified load identification model, combined with the use of a grid-insensitive structural stress method to evaluate the life-span of the boom weld structure, which can potentially reflect the actual failure location of the weld. Xiang et al. (2017) utilized load spectrum testing methods to indirectly perceive loading forces exerted by excavators. Moreover, Ilaria et al. (2019) applied regularization methods to accurately estimate real-time digging forces and loads exerted on excavator buckets. Additionally, Gu et al. (2023) proposed a multifunctional proportional control triaxial stress loading device, which can apply output force to coal rock to simulate triaxial stress loading conditions. Data-driven indirect force perception methods use detection signals to extract features based on time, frequency, or envelope spectrum characteristics. These methodologies employ signal processing, data mining, modern artificial intelligence, and other technologies to detect and diagnose variations in applied forces (Eraliev et al. 2022). Yang et al. (2021) introduced a time-domain extrapolation method based on Markov chain Monte Carlo to predict peak loads surpassing a threshold on tractor transmission shafts, effectively capturing the time frequency characteristics of the load. Similarly, Wang et al. (2023) introduced a torque soft measurement method to handle extensive data from loader transmission shafts, employing a distributed sparse least-squares support vector machine algorithm to accurately characterize the torque under loader operating conditions. While data-driven methods are gaining traction in technological development, their stability in engineering applications necessitates thorough examination. In contrast, model-based indirect force perception methods rely on constructing mechanical models of mechanisms and utilizing information from detection points to estimate the forces acting on the end of the mechanism. This approach provides valuable insights for our study.

The clamping mechanism of forging manipulators holds inherent advantages for indirect perception of forging forces due to its closer proximity to the high-temperature forged parts. Therefore, this article proposes to investigate indirect perception methods for forging forces utilizing the clamping mechanism. Lanzhou Jiaotong University and Central South University have conducted relatively comprehensive research on the mechanical modeling of the clamping mechanism (Li, Gao, and Deng 2010; Yang, Zhang, and Jiang 2014; Yang et al. 2017; Ding et al. 2012; He et al. 2009). However, research has revealed that the aforementioned mechanical models cannot fully meet the indirect perception requirements for forging forces. This limitation arises because the mentioned models primarily concentrate on force analysis and research under extreme positions (clamps in vertical and horizontal states), thereby neglecting to account for variations in the attitude of the clamp and the consequential impact of such variations on the force state of the clamping mechanism. Therefore, this study aims to develop a mechanical model that integrates variables related to the attitude of the clamping mechanism, thereby providing a

quantitative depiction of the forging forces. Through experimental investigations, the feasibility of indirectly perceiving forging forces will be validated, and the perception law will be analyzed. The objective is to furnish a theoretical foundation and technical assistance for the adaptable control of forging manipulators.

The subsequent sections of this article are structured as follows: [Section 2](#) delineates the construction of the full attitude force analytical model and hinge-point force model of the clamping mechanism; [Section 3](#) presents the proposed method for perceiving forging forces; [Section 4](#) scrutinizes the indirect perception capability and law of forging forces; and, lastly, [Section 5](#) offers a summary and discussion of the innovations introduced in this paper.

2. Force flow analysis and mechanical modeling

2.1. Analysis of force transmission paths

The forging manipulator involved in this research is a 20 kN rail hydraulic forging manipulator in 5 MN fast forging hydraulic unit pilot test platform of Yanshan University. The composition of the clamping mechanism of the forging manipulator is shown in [Fig. 2](#). The piston rod of the hydraulic cylinder is attached to the forging manipulator, while the remaining components are connected via pin shaft hinges. Power is transmitted between components through the drive cylinder barrel, enabling the clamping and manipulation of the forging by the jaws.

In exploring the indirect perception of forging forces, it is essential to clarify the transmission path of these forces. When the forging undergoes deformation due to the force exerted by the press, both the axial and vertical components of the forging force act simultaneously on the clamp jaw. Subsequently, they pass through the hinge pin shafts, reaching the clamp arm, clamp shell, and connecting rod before ultimately reaching the hydraulic cylinder, as illustrated in [Fig. 2](#). It is evident that, as the sole clamping object of the forging, the transmission path of forces within the clamping mechanism remains unchanged, and the mode of action of the axial component of the forging force in the coordinate system also remains unchanged. However, considering that the clamping mechanism rotates continuously during operation, its attitude and the vertical component of the forging force it receives vary. Therefore, constructing a full-attitude mechanical model of the clamping mechanism becomes necessary to perceive the forging force under different attitudes.

Throughout the full rotation of the clamping mechanism, the vertical and horizontal clamping of the jaws on the forging represent two extreme positions of the clamping status. The remaining clamping attitudes are transitional states in which the two extreme positions of clamping alternate with each other. These transitional states are referred to as general clamping states. The forging forces in extreme positions are relatively easy to analyze mechanically due to the special direction of force. However, in general clamping states, factors, such as tilting of the forging and friction lead to much more complex mechanical analysis with limited accuracy. Therefore, for general clamping states, this study employs the concept of orthogonal decomposition for analysis. In the coordinate system of the clamping mechanism, the forging force is decomposed into the vertical

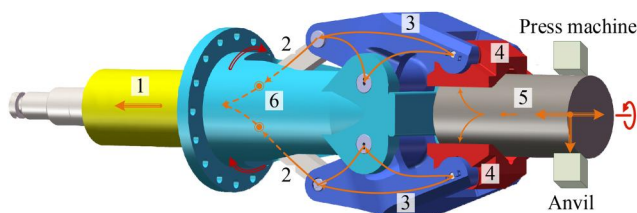


Figure 2. The clamping mechanism and its force transmission paths (1. hydraulic cylinder; 2. connecting rod; 3. clamp arm; 4. clamp jaw; 5. forging; 6. clamp shell.).

clamping force, F_v , and the horizontal clamping force, F_h , as illustrated in Fig. 3. It can be observed that F_v is perpendicular to the jaw, while F_h is parallel to the jaw. To solve the forging force in any attitude, it can be decomposed into F_v and F_h for calculation, thereby achieving a full-attitude mechanical description of the forging force.

2.2. Full-attitude mechanical modeling

In the forging process, the full rotation of the clamping mechanism adjusts the attitude of the forging. The rotation angle of the clamping mechanism and the opening angle of the clamp jaw are the main variables determining the clamping attitude. It is necessary to comprehensively describe any clamping position. φ is defined to represent the rotation angle of the clamping mechanism, as shown in Fig. 3.

Figure 4 shows a schematic diagram of the clamping mechanism of the forging manipulator, and the main parameters of the corresponding components are listed in Table 1. S_1 to S_3 indicate the centroid positions of various components, while A, B, C, and O represent the hinge-points between the components. d represents the equivalent diameter of the forging clamping end. α , β , and θ denote the angles between each component and the horizontal direction, while δ is the angle between the clamp jaw, h_O is the distance between hinge-point O and the symmetry line, h_A is the distance between hinge-point O and hinge-point A, h_C is the distance between hinge-point C and the symmetry line, and γ represents the angle between the line connecting hinge-point O-C and the line connecting S_3 to hinge-point O. Y represents the distance from hinge-point C to the tip of the angle of the clamp jaw. When the equivalent diameter d , varies, the opening angle of the clamp jaw changes, and α , β , and θ also change accordingly. Therefore, α , β , and θ are used to represent the attitude variables of the clamping mechanism, yielding the following results:

$$\gamma = \arccos \frac{l_{OC}^2 + l_{OS_3}^2 - l_{CS_3}^2}{2 \cdot l_{OC} \cdot l_{OS_3}} \quad (1)$$

$$\beta = \arcsin \left[\frac{d/2 \sin(\delta/2) + Y - h_O}{l_{OC}} \right] \quad (2)$$

Hence, the expression for θ is obtained as follows:

$$\theta = \pi - \beta - \angle BOC \quad (3)$$

Finally, the expression for α is obtained as follows:

$$\alpha = \arcsin \left(\frac{l_{OB} \cdot \sin \theta + h_{OA}}{l_{AB}} \right) \quad (4)$$

The angle α , which incorporates β , θ , and the basic parameters of each clamp component, provides a comprehensive expression of the positional variations of the connecting rod, clamp arm,

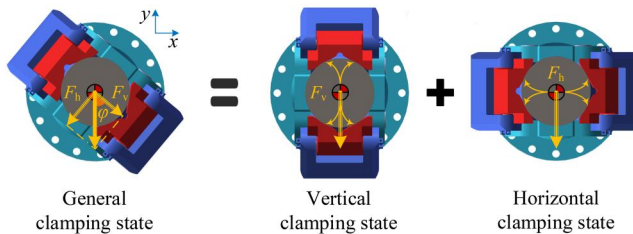


Figure 3. Rotational molding process flow chart.

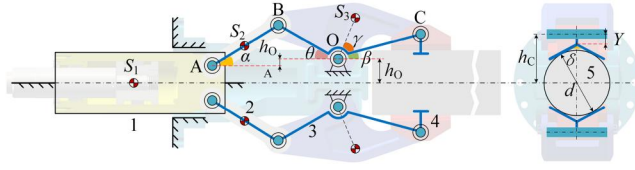


Figure 4. Structural schematic of the forging manipulator's clamping mechanism.

Table 1. Main parameters of the clamping mechanism components

Parameter type	Parameter value
Mass of connecting rod 2 (m_2)	16.7 kg
Length of connecting rod 2 (l_2)	360mm
Mass of clamp arm 3 (m_3)	178.4 kg
Distance between hinge points O and B (l_{OB})	320 mm
Distance between hinge points O and C (l_{OC})	400 mm
Vertical distance from hinge point A to O (h_{OA})	30 mm
Distance from hinge point O to the central plane (h_O)	110 mm
Mass of clamp jaw (m_4)	54.3 kg
Angle of clamp jaw (δ)	120°
Distance from hinge point C to the vertex of the jaw clamping angle (Y)	47.7 mm

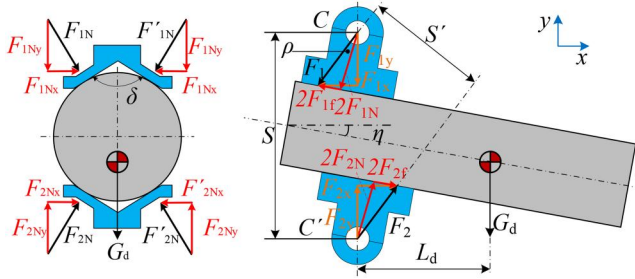


Figure 5. Mechanical analysis of the vertical clamping state of the clamping mechanism.

and jaw under different jaw opening angles through Eqs. (1)–(4). This expression centers on α as the parameter closest to the clamping cylinder, effectively capturing the pose changes in the system.

According to the concept of orthogonal decomposition of forces, any general clamping state at an arbitrary angle, φ , is defined as the vector sum of the vertical clamping state component and the horizontal clamping state component. Using vector methods, a full-attitude forging force model is established. First, the force models of the vertical clamping state and the horizontal clamping state are solved through mechanical analysis.

2.2.1. Force modeling of vertical clamping state

The mechanical analysis of the vertical clamping state of the clamping mechanism is depicted in Fig. 5. Two clamps apply clamping forces F_1 and F_2 to the forging to overcome the forging force. The downward inclination angle, η , induced by the forging force and gravity on the forging is defined. During forging, it can be obtained through the angular displacement sensor at hinge-point C. Considering the effect of friction, F_1 and F_2 are decomposed into the normal force, F_N , perpendicular to the surface of the forging and the friction force, F_f , along the axis of the forging. Additionally, F_N is decomposed into F_{Nx} and F_{Ny} along the x and y directions, respectively. In consideration of the influence of gravity, G_d , on the forgings, from the mechanical equilibrium and geometric relationships, the system of equations is obtained as follows:

$$\begin{cases} G_d \cdot L_d = S' \cdot F_1 \\ S' = S \cdot \sin(\rho + \eta) \\ F_{1y} = F_1 \cdot \cos(\rho + \eta) \\ F_{2y} = F_{1y} + G_d \\ \rho = \arctan \mu \end{cases} \quad (5)$$

where ρ is the friction angle between the forging and the clamp; μ is the friction coefficient between the forging piece and the clamp; L_d represents the distance from the forging center of gravity to the center of the jaw pin. S' is the vertical distance from the center of the lower jaw pin to the line of action of the resultant force; S denotes the distance between the centers of the upper and lower jaw pins; and η is the permissible rotational angle for the forging piece to sag, with a typical range of $\eta = 0 \sim 4^\circ$.

It can be obtained from Eq. (5):

$$F_1 = \frac{G_d \cdot L_d}{S'} = \frac{G_d \cdot L_d}{S \cdot \sin(\rho + \eta)} \quad (6)$$

The vertical components of the resultant force F_1 , denoted as F_{1y} and F_{2y} , are:

$$\begin{cases} F_{1y} = F_1 \cdot \cos(\rho + \eta) = \frac{G_d \cdot L_d}{S \cdot \tan(\rho + \eta)} \\ F_{2y} = F_1 + G_d = G_d \cdot \left[1 + \frac{L_d}{S \cdot \tan(\rho + \eta)} \right] \end{cases} \quad (7)$$

Define F_{J1} as the total clamping force. From Fig. 5, the total clamping force of the upper and lower jaws is

$$F_{J1} = F_{1y} + F_{2y} = G_d \cdot \left[1 + \frac{2L_d}{S \cdot \tan(\rho + \eta)} \right] \quad (8)$$

When the jaws are in a vertical position, the output force of the hydraulic cylinder is obtained as:

$$F_{J1} = \frac{k\zeta}{i} (F_{1y} + F_{2y}) = G_d \cdot \frac{k\zeta}{i} \cdot \left[1 + \frac{2L_d}{S \cdot \tan(\rho + \eta)} \right] \quad (9)$$

where k represents the tightening force storage coefficient, usually taken as 1.2; ζ denotes the efficiency of the clamping mechanism, typically taken as 0.8; and i stands for the force arm ratio of the clamp arm, $i = \frac{l_{OB} \cdot \sin(\pi - \alpha - \theta)}{l_{OC} \cdot \cos \beta \cos \alpha}$.

2.2.2. Force modeling of horizontal clamping state

The horizontal clamping state of the clamping mechanism is when the clamp is positioned on both sides of the forging, as shown in Fig. 6. The symmetric parts experience the same forces,

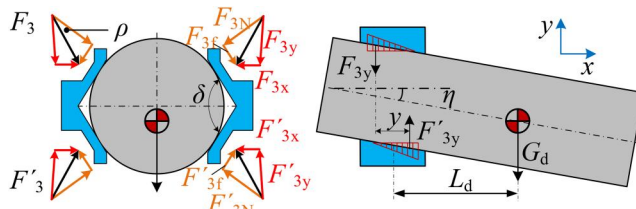


Figure 6. Mechanical analysis of the horizontal clamping state of the clamping mechanism.

with a downward inclination angle of η . Based on the principles of mechanical equilibrium and geometric relationships, the equation set can be obtained as follows:

$$\begin{cases} 2F_{3y} \cdot y - G_d \cdot (L_d - y/2) = 0 \\ 2F'_{3y} \cdot y - G_d \cdot (L_d + y/2) = 0 \end{cases} \quad (10)$$

where, y is the axial distance between the moments of the two jaws, and its value depends on the distribution of the moment along the jaw length l_j . When the positive moment is distributed in a triangular shape along the jaw length, $y=l_j/3$; when the moment is distributed in a triangular shape over half the jaw length, y can be taken as $y=2l_j/3$. For calculation purposes, y can be approximated as $y=l_j/2$.

It can be obtained from Eq. (10):

$$\begin{cases} F_{3y} = G_d \cdot \frac{2L_d - y}{4y} \\ F'_{3y} = G_d \cdot \frac{2L_d + y}{4y} \end{cases} \quad (11)$$

The horizontal components of the total reaction forces F_3 and F'_3 are denoted as F_{3x} and F'_{3x} , respectively. From Fig. 6, we can obtain:

$$\begin{cases} F_{3x} = \frac{2L_d - y}{4y} \cdot G_d \tan\left(\frac{\delta}{2} - \rho\right) \\ F'_{3x} = \frac{2L_d + y}{4y} \cdot G_d \tan\left(\frac{\delta}{2} - \rho\right) \end{cases} \quad (12)$$

The total clamping force is defined as F_{j2} , and its expression is as follows

$$F_{j2} = 2(F_{3x} + F'_{3x}) = \frac{2L_d}{y} \cdot G_d \tan\left(\frac{\delta}{2} - \rho\right) \quad (13)$$

When the jaws are in the horizontal position, the output force of the hydraulic cylinder can be obtained as:

$$F_{j2} = \frac{k\zeta}{i} \cdot 2(F_{3x} + F'_{3x}) = \frac{2k\zeta}{i} \cdot \frac{G_d L_d}{y} \cdot \tan\left(\frac{\delta}{2} - \rho\right) \quad (14)$$

In summary, define F_j as the thrust required for clamping the forging by the hydraulic cylinder in the general state. By vectorially summing F_{j1} and F_{j2} , combined with Eqs. (9) and (14), the mapping model of the hydraulic cylinder thrust, F_j , with clamping attitude and forging force can be obtained as follows:

$$F_j = \frac{2k\zeta G_d L_d}{i} \cdot \sqrt{\left[\frac{\cos \varphi}{S \cdot \tan(\rho + \eta)} + 1\right]^2 + \left[\frac{\tan(\delta/2 - \rho) \cdot \sin \varphi}{y}\right]^2} \quad (15)$$

2.2.3. Analysis of the impact of changes in frictional states

To simplify the model and emphasize the core principles of the study, a constant friction coefficient is assumed to reflect the mechanical characteristics under typical operating conditions. However, the dynamic variation of the friction coefficient may affect the model's accuracy and

applicability. To address this, we have conducted the following specific analysis regarding the potential impact of dynamic friction coefficients on the model:

Impact of friction coefficient on the model

The magnitude of frictional force is directly controlled by the friction coefficient μ and the supporting force F_N . Changes in the friction coefficient can lead to errors in the force distribution at the hinge point, especially in the direction along the contact surface. Additionally, errors in multi-hinge systems may accumulate, causing nonlinear effects on the overall perception capability evaluation.

Causes of friction coefficient variation

The friction coefficient can change dynamically during actual forging operations due to various factors, including:

- Material properties (such as surface roughness and hardness) changing due to processing or wear.
- Environmental conditions (such as high temperatures leading to the decomposition or change of lubricant properties).
- Mechanical states (such as wear or deformation of the clamping mechanism leading to changes in contact characteristics).
- Vibrations or impacts causing transient contact changes.

2.3. Establishing the mapping relationship between hinge-points and forging forces

The clamping mechanism has inherent advantages in the indirect perception of forging forces. Hinge-points, as the connecting points between various components of the clamping mechanism, are crucial for the indirect perception of forging forces. Therefore, establishing a force mapping model between each hinge-point and the forging forces is essential for characterizing forging forces through hinge-point forces. Firstly, the force analysis of the components is conducted separately. Figure 7 shows a mechanical analysis of the hydraulic cylinder. The hinge-point forces at A and A' are orthogonally decomposed into the x-direction and the y-direction. Combined with Eq. (15), the x-component F_{21x} and F'_{21x} of the hinge-point forces at A and A' can be obtained as follows:

$$\begin{cases} F_{21x} = \sqrt{\left(\frac{k\zeta}{i} F_{1y} \cdot \cos \varphi\right)^2 + \left(\frac{F_J}{2} \cdot \sin \varphi\right)^2} \\ F'_{21x} = \sqrt{\left(\frac{k\zeta}{i} F_{2y} \cdot \cos \varphi\right)^2 + \left(\frac{F_J}{2} \cdot \sin \varphi\right)^2} \end{cases} \quad (16)$$

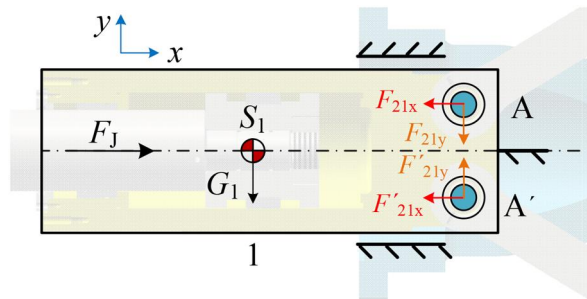


Figure 7. Mechanical analysis of the hydraulic cylinder.

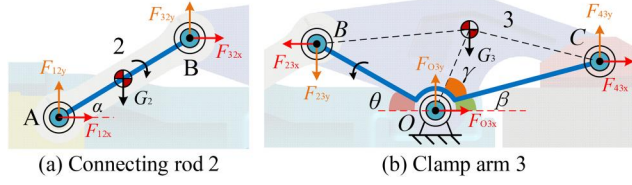


Figure 8. The mechanical analysis of the connecting rod 2 and the clamp arm 3.

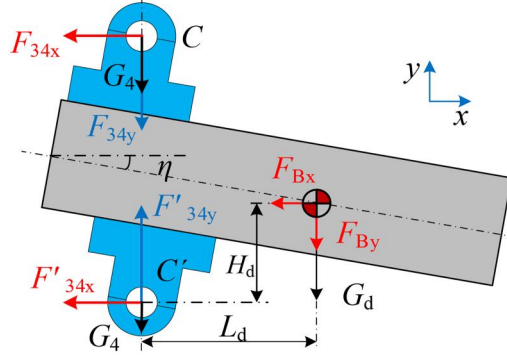


Figure 9. Mechanical analysis of the clamp jaw and forging.

The mechanical analysis of the connecting rod is shown in Fig. 8(a). Considering that friction issues involved in this study exist at various hinge-points in the form of friction torque and that the friction form is relatively simple, we chose the Coulomb friction model from among many friction models to consider frictional forces (Liu et al. 2015). Based on force and torque balance equations, the equation set can be formulated as follows:

$$\begin{cases} F_{12x} + F_{32x} = 0 \\ F_{12y} + F_{32y} - G_2 \cdot \cos \varphi = 0 \\ F_{12x} \cdot \sin \alpha \cdot \frac{l_{AB}}{2} - F_{32x} \cdot \sin \alpha \cdot \frac{l_{AB}}{2} - F_{12y} \cdot \cos \alpha \cdot \frac{l_{AB}}{2} + F_{32y} \cdot \cos \alpha \cdot \frac{l_{AB}}{2} + M_{12f} + M_{32f} = 0 \\ M_{12f} = \rho_A \cdot \sqrt{F_{12x}^2 + F_{12y}^2} \\ M_{32f} = \rho_B \cdot \sqrt{F_{32x}^2 + F_{32y}^2} \end{cases} \quad (17)$$

From Eq. (16), we obtain F_{12x} . Given F_{12x} , we can solve for F_{12y} , F_{32x} , and F_{32y} . Then, we can obtain the orthogonal hinge forces at points A and B. The mechanical analysis of the clamp arm, clamp jaw, and forging is shown in Figs. 8(b) and 9. Similarly, equations are derived based on the principles of force and torque equilibrium as follows:

$$\begin{cases} F_{34x} + F'_{34x} + F_{Bx} = 0 \\ -F_{34y} + F'_{34y} - (2G_4 \cdot \cos \varphi + G_d \cdot \cos \varphi + F_{By}) = 0 \\ 2F_{34x} \cdot h_C + F_{Bx} \cdot H_d - L_d \cdot (F_{By} + G_d \cdot \cos \varphi) + M_{Cf} - M'_{Cf} = 0 \\ M_{Cf} = \rho_C \cdot \sqrt{F_{34x}^2 + F_{34y}^2} \\ M'_{Cf} = \rho_C \cdot \sqrt{F'_{34x}^2 + F'_{34y}^2} \end{cases} \quad (18)$$

$$\begin{cases}
-F_{23x} + F_{O3x} + F_{43x} = 0 \\
-F_{23y} + F_{O3y} + F_{43y} - G_3 \cdot \cos \varphi = 0 \\
F_{23x} \cdot \sin \theta \cdot l_{OB} + F_{23y} \cdot \cos \theta \cdot l_{OB} - \\
G_3 \cdot l_{OS3} \cdot \cos \varphi \cdot \cos (\beta + \gamma) + F_{43y} \cdot \cos \beta \cdot l_{OC} - \\
F_{43x} \cdot \sin \beta \cdot l_{OC} - M_{23f} - M_{O3f} - M_{43f} = 0 \\
M_{O3f} = \rho_O \cdot \sqrt{F_{O3x}^2 + F_{O3y}^2} \\
M_{43f} = \rho_C \cdot \sqrt{F_{43x}^2 + F_{43y}^2}
\end{cases} \quad (19)$$

By combining Eqs. (17) and (19), the orthogonal hinge forces at hinge-points O and C can be solved. Therefore, Eqs. (17)–(20) establish the mapping model between the hinge forces at various hinge-points of the clamping mechanism and the forging force. Through these hinge-points, indirect perception of the forging force can be achieved.

3. Indirect perception methods and experiments

3.1. Experimental platform and indirect perception methods

To verify the feasibility of the indirect perception method for the forging force of the clamping mechanism and the effectiveness of the full-attitude force model and to obtain data from each hinge-point for analyzing their capabilities and laws in relation to perception of forging force, this study constructed a clamping mechanism experimental platform and proposed a testing method. The experimental platform, as shown in Fig. 10, consisted of a scaled-down prototype of the clamping mechanism (scale-down ratio: 1:5), a hydraulic loading system, and a strain measurement system. The details of the experimental platform are provided in Table 2. This study employed the orthogonal decomposition approach to construct a full-attitude force perception model. The vertical clamping state and horizontal clamping state are crucial for perceiving the general clamping state of the forging force through the model. Therefore, the clamping mechanism prototype was installed on a reversible mounting bracket, supporting the prototype to complete tests in both vertical and horizontal clamping states.

Additionally, to provide a clearer description in the next section regarding the perception capabilities and laws of different hinge-points with respect to the forging force, the hinge-points of the clamping mechanism are classified. The hinge-points in the upper part of Fig. 10 are defined as upper hinge-points, while those in the lower part are defined as lower hinge-points.

To detect the change in the hinge force, by means of which the forging force is detected, this study employed a resistance strain gauge testing method. Mechanical data were collected using

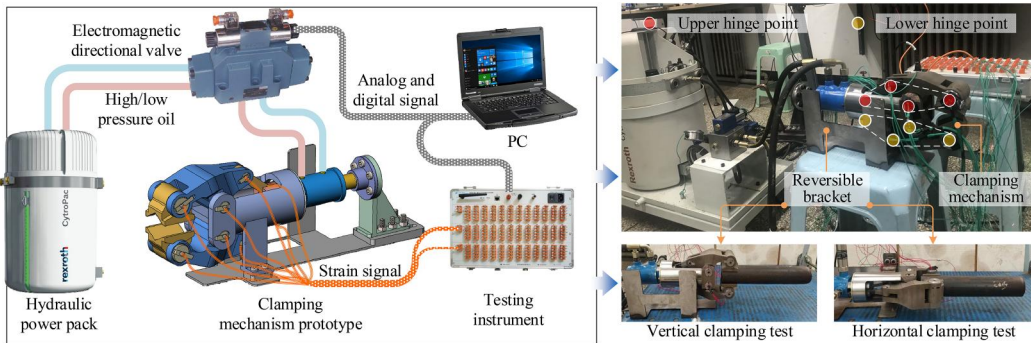
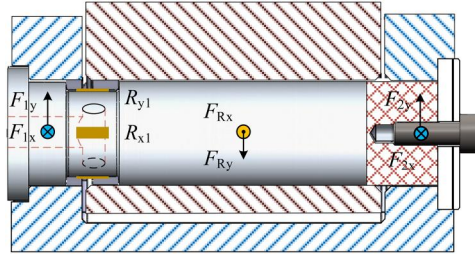


Figure 10. Experimental platform for indirect perception of forging force in clamping mechanism.

Table 2. Experimental platform component details table

Name	Model	Specification
Hydraulic power pack	Rexroth CytoPac	Power: 3 kW, Maximum Pressure: 240 bar, Rated Flow Rate: 160 L/min
Directional valve	Rexroth 4WE6J	/
Strain gauge	BX120-0.5AA	Sensitivity coefficient: $2.08 \pm 1\%$, Resistance value: $120 \pm 0.2 \Omega$
Data acquisition system	DH3816N	3 main channels, 36 measurement points

**Figure 11.** Layout diagram of strain gauges installed inside the hinge pin sleeve.

strain gauges through a strain gauge instrument. Currently, high-temperature strain gauges are available on the market that are suitable for the working environment of forging machines and capable of wireless data collection. However, considering the feasibility of the indirect perception method for assessing the forging force of the forging manipulator in this study, to reduce testing costs, the experiment was conducted at normal temperature, and wired metal foil resistance strain gauges were selected for the study. The layout of the strain gauges inside the hinge pin is depicted in Fig. 11. Due to the limited space within the holes of the hinge pins of the prototype, it was challenging to arrange the strain gauges and leads. Therefore, the hinge pins of the prototype were modified to create grooves for attaching the strain gauges and holes for the leads. The strain gauges were arranged in the x and y-directions of the hinge pin to construct a one-quarter bridge test circuit.

3.2. Experiment and results analysis

The prototype simulated vertical and horizontal clamping states with a $\varnothing 70 \text{ mm} \times 400 \text{ mm}$ steel rod. The pressure values of two chambers of the hydraulic cylinder were measured to obtain the hydraulic cylinder thrust and the strain values of each hinge-point. The pressure values of the hydraulic cylinder and the calculated thrust values are shown in Fig. 12. It can be observed that the pressure in the rod and the rodless chambers of the hydraulic cylinder remains constant, indicating stable thrust values. By substituting the thrust of the hydraulic cylinder into Eqs. (16)–(20), the numerical solutions of the hinge forces at each hinge-point were obtained and compared with the experimentally obtained results, as shown in Fig. 13.

Figure 13 shows that under both vertical clamping and horizontal clamping states, the hinge point force test values of the experimental prototype are generally consistent with the theoretical values from the established hinge point force analytical model. The test values at each hinge point are slightly smaller than the theoretical analytical values. Among them, the deviations at hinge points A and B are relatively large, with the maximum deviation being 9.3%. The analysis suggests that the main sources of error include the following aspects:

1. Hydraulic cylinder processing errors and friction: The hydraulic cylinder in the clamping mechanism prototype has certain processing errors, leading to smaller fitting gaps and

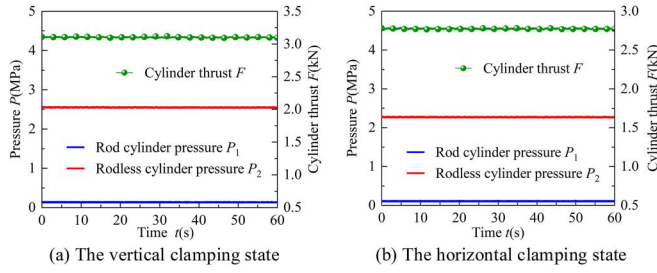


Figure 12. The hydraulic cylinder thrust in different clamping states.

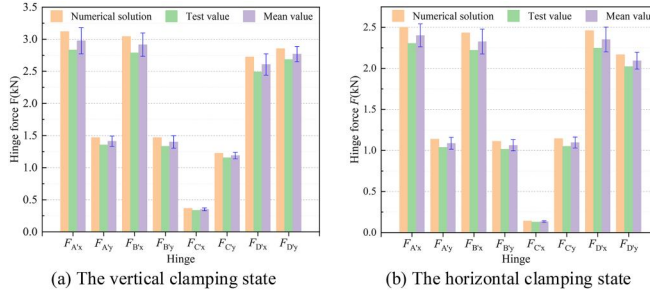


Figure 13. Quantitative analysis of the indirect perception experiment of forging force.

increased friction, which causes the test values to be slightly lower than the theoretical values.

2. Influence of support contact state: The support state between the shaft and component is a line contact envelope, which partly restricts the free deformation of the pin and hinders the force transmission, leading to experimental data lower than the theoretical values.
3. Hydraulic system and sensor fluctuations: The hydraulic loading system may experience pressure fluctuations during operation, and the voltage signal of the pressure sensor may also fluctuate, leading to instability in the hydraulic cylinder output force, contributing to errors.

Additionally, under extreme working conditions such as high temperature or heavy load, there may be additional sources of error, such as:

- Sensor accuracy decline: Temperature changes may impair the accuracy of the pressure sensor, affecting force measurement precision.
- Hydraulic system stability decrease: Heavy load conditions may increase the amplitude of pressure fluctuations in the hydraulic system, further interfering with the precision of force output.
- Structural Deformation: High-temperature conditions may cause a decrease in the material rigidity of the clamping mechanism, resulting in greater deformation and affecting the force transmission pattern.

In conclusion, although some factors may affect the test results, these errors are within an acceptable range in the overall model results. This proves that the force perception method based on the hinge point forces of the clamping mechanism proposed in this study is feasible within reasonable error margins and can effectively reproduce the dynamic changes of the hinge point forces under periodic rotational combined movements. The established full-posture forging force perception model can accurately describe the variation patterns of hinge point forces under typical clamping states, especially when the clamping mechanism posture changes (such as rotation

or opening and closing), demonstrating good applicability in modeling the dynamic characteristics of hinge point forces.

Compared to existing studies, which primarily focus on external load identification (Yuan et al. 2019; Wang et al. 2023), load spectrum testing (Xiang et al. 2017; Ilaria et al. 2019), or fatigue life assessment (Yang et al. 2021), while providing valuable analytical tools, these studies often fail to fully consider the profound impact of posture changes on mechanical behavior. As a result, they have certain limitations when dealing with complex working conditions in engineering applications. This study represents an advancement in the innovation of posture change and force characteristic point sensitivity analysis within the force perception model. The innovation of this study lies in its thorough exploration of the relationship between the posture variations of the clamping mechanism and the mechanical characteristic points, which enhances the model's adaptability.

4. Analysis of the perception capabilities and laws of hinge-points

The clamping mechanism undergoes periodic rotation during the clamping of the forging, resulting in differences in the force perception capability and patterns at various hinge-points for different types of forging forces. Therefore, based on the validated full-attitude forging force indirect perception model, numerical simulations are conducted for multiple operating conditions. This study investigates the perception capabilities and patterns of forging forces at each hinge-point under different conditions.

4.1. Change in the laws of hinge-point forces with a clamping attitude

The clamping mechanism undergoes rotation throughout the forging process, resulting in periodic fluctuations in the hinge-point forces, consequently influencing their perception capabilities. Therefore, to simulate the variation in hinge-point forces with clamping attitude, we shall assume that both the axial and vertical components of the forging force are 50 N and analyze the perception capabilities and laws of each hinge-point. By employing Eqs. (16)–(20), we can derive the relationship between hinge-point forces and the rotation angle of the clamping mechanism, as depicted in Fig. 14. During the full rotation process, the positions of the upper and lower hinge-points defined in Fig. 10 alternate, implying that the perception capabilities and trends remain consistent. Therefore, we only need to analyze the upper hinge-point. In the figure, the direction of the force is represented by positive and negative values.

From Fig. 14, it can be observed that the fluctuation range of the x-axis component of the hinge-point force in Fig. 14(a) is larger compared to the y-axis component in Fig. 14(b). A broader fluctuation range signifies greater sensitivity to changes in clamping mechanism rotation

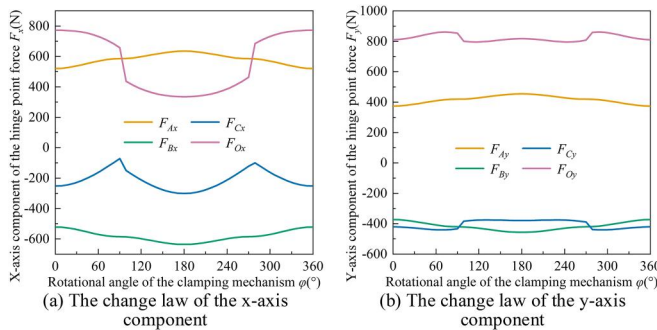


Figure 14. The change law of the hinge-point force at the upper hinge-point of the clamping mechanism.

within the same interval. Therefore, the perception capability of the x-axis component of the hinge-point force is stronger compared to the y-axis component, making it more suitable as a perception parameter for clamping mechanism rotation. Within the x-axis component of hinge-point force, it can be seen that the curve curvature of hinge-point C is the largest, followed by that of hinge-point O, while hinge-points A and B have the smallest curvatures. A larger curvature indicates greater perceptivity with respect to clamping mechanism rotation changes. Thus, hinge-points O and C exhibit the strongest perception capabilities concerning alterations in clamping attitude. In summary, the x-axis components of the hinge-point forces of hinge-points O, O', C, and C' can be utilized for perceiving clamping attitude.

4.2. Change in the laws of hinge-point forces with forging diameter

In the forging process, the diameter and centroid of the forged piece change as a result of forging and elongation, which is a typical scenario in free forging. The hinge-point forces perceive changes in the diameter and centroid of the forged piece, aiding in process adjustments during forging. Simulating the variation in hinge-point forces with the diameter of the forged piece, based on the premise of vertically clamping a constant mass forged piece with the clamping mechanism, involves solving for the required driving force using Eq. (9) and obtaining the hinge forces of each hinge-point through Eqs. (16)–(20). Assuming a constant mass of 1000 kg for the forged piece, with axial and vertical force components both experiencing 5000 N of force, based on the effective clamping range of the clamping mechanism, the diameter of the forged piece at the clamping end is set to 300 mm and remains constant. The forged piece diameter is treated as a single variable, with the diameter variation in the forged and hammered section ranging from 100 to 500 mm. The influence of forged piece diameter change on the hinge-point forces of the upper and lower sections is illustrated in Fig. 15, where the direction of the forces is indicated by positive and negative values. Since the diameter reduction process is characteristic of the elongation process, the x-axis diameter change in the Figures shows a decreasing trend.

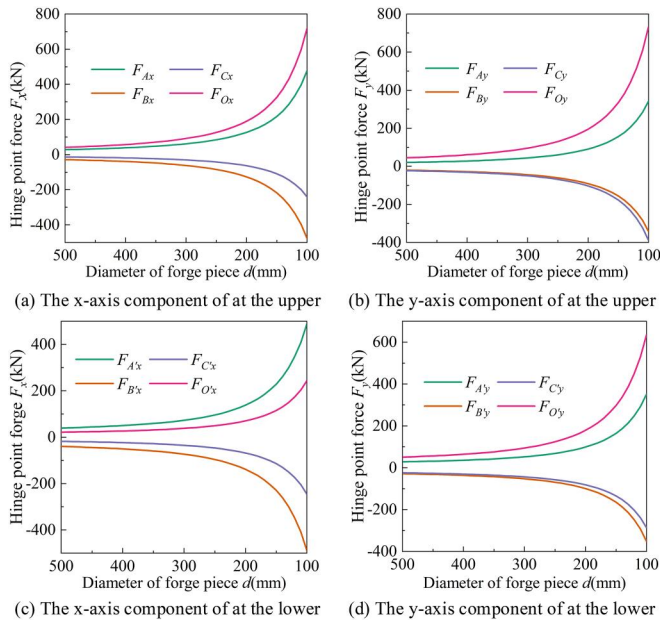


Figure 15. The stress law of hinge-point of clamping mechanism varies with forging diameter.

The overall analysis of Fig. 15 reveals that the x-axis and y-axis components of the hinge-point force at both the upper and lower hinge-points show an increasing trend as the diameter decreases, with the slope of the curves gradually increasing. A larger change in the slope of the hinge-point force variation curve indicates a stronger perceptivity of the hinge-point force with respect to changes in the diameter of the forging. Therefore, the perception trend of all hinge-points is consistent. Further comparison of the slope changes in the curves reveals the following:

1. In Fig. 15(a), the slope change of the x-axis component of the upper hinge-point O is the greatest, followed by hinge-points A and B, with hinge-point C exhibiting the poorest performance. Comparing the variation in the hinge-point force in the y-axis component shown in Fig. 15(b), the slope change of the y-axis component of hinge-point O remains the largest, followed by hinge-point C, whose slope change is slightly higher than that of hinge-points A and B. These characteristics indicate that, among the upper hinge-points, hinge-point O exhibits the strongest perceptivity to changes in the forging diameter.
2. In Fig. 15(c,d), among the lower hinge-points, hinge-points A' and B' exhibit the best performance in terms of the x-axis component, while hinge-point C' slightly outperforms O'. Regarding the y-axis component, hinge-point O' demonstrates the best performance, followed by hinge-points A' and B', with hinge-point C' exhibiting the poorest performance.

Overall, it can be observed that hinge-points A, A', B, and B' generally exhibit moderate perception capabilities for the diameter changes induced by the forging process. Hinge-points O and O', on the other hand, demonstrate the optimal perception capabilities for the y-axis component of the hinge-point force. Throughout the forging process, the diameter changes in the forging can be sensed through the hinge-points O and O' of the clamping mechanism.

4.3. Change in the laws of hinge-point forces with fluctuations in forging force

4.2.1. The axial component of forging force changes

During the forging process, due to the downward pressure of the press or double-sided constraints, the axial component of the forging force undergoes significant changes, leading to intense horizontal impacts on the forging manipulator. A clear perception of the axial component is beneficial for providing timely buffering against these impacts. A numerical simulation of the variation in the axial component of the forging force was conducted, in which it was treated as the sole variable. The range of variation was set from -100 to 100 N based on the order of magnitude, while the vertical component of the forging force remained constant at 50 N. The law of the hinge-point force in response to the variation in the axial component of the forging force was obtained using Eqs. (16)–(20), as illustrated in Fig. 16.

Figure 16 illustrates that as the axial component of the forging force varies, the hinge-point forces of A, A', B, and B' demonstrate minimal fluctuations, with their curves being nearly horizontal, indicating negligible perceptivity from a perceptual standpoint. Conversely, the hinge-point forces of O, O', C, and C' exhibit fluctuations in response to changes in the axial component of the forging force. Further analysis of the slope of these curves indicates that the x-axis component of the hinge forces at the lower hinge-points O' and C' exhibits a larger slope compared to the y-axis component, indicating a heightened sensitivity of the x-axis component to variations in the axial component of the forging force. Therefore, the x-axis components of the hinge forces at the lower hinge-points O' and C', designated as $F_{O'x}$ and $F_{C'x}$, respectively, can be selected as sensitive hinge-points for perceiving fluctuations and impacts in the axial component of the forging force.

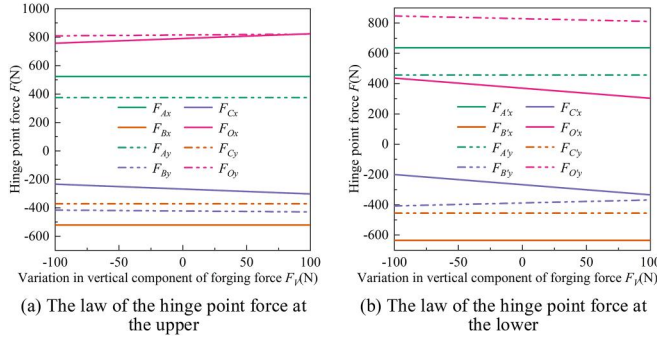


Figure 16. The stress law of the hinge-point of the clamping mechanism with the change of the axial component of the forging force.

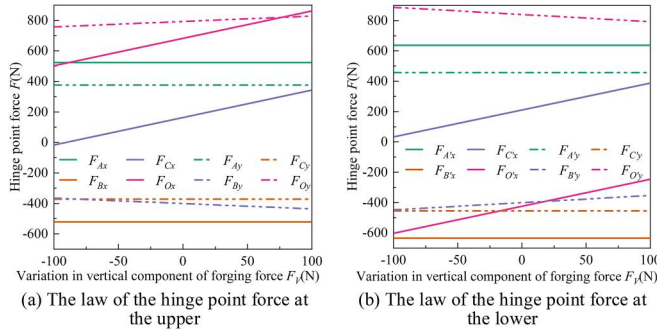


Figure 17. The stress law of the hinge-point of clamping mechanism with the change of vertical component of forging force.

4.2.2. The vertical component of forging force changes

During the forging process, the downward pressure from the press exerts a significant vertical impact on the forging force, leading to notable fluctuations in the vertical component of the forging force. A numerical simulation of the vertical component of the forging force was conducted to analyze the perception capability of each hinge-point. By considering the vertical component of the forging force as a single variable within a predefined range of -100 to 100 N, while maintaining the axial component constant at 50 N, the relationship between the hinge-point force and the vertical component of the forging force was obtained through Eqs. (16)–(20), as shown in Fig. 17.

From Fig. 17, it can be observed that the hinge-point forces of A, A', B, and B' exhibit minimal fluctuations with the change in the vertical component of the forging force, indicating a lack of perception capability. However, the hinge-point forces of O, O', C, and C' exhibit noticeable variations with changes in the vertical component of the forging force. The further comparison reveals that the slope of the hinge-point forces x-axis components for O, O', C, and C' is larger than that of the y-axis components, suggesting a stronger perception capability for the vertical component of the forging force. Therefore, the x-axis components F_{Cx} , $F_{C'x}$, F_{Ox} , and $F_{O'x}$ of the hinge-point forces of O, O', C, and C' can be chosen as sensitive hinge-points for sensing the fluctuations and impacts of the vertical component of the forging force.

5. Conclusions

The research findings and conclusions of this study are as follows:

1. A full-attitude indirect perception model for forging forces based on the hinge-point forces of the clamping mechanism is proposed. The model considers state variables such as the

clamping posture of the mechanism and the equivalent diameter of the forging. It provides a mathematical description of the hinge-point forces of the clamping mechanism under all postures, establishing a mapping relationship between the forging forces and the hinge-point forces of the clamping mechanism.

2. The perception capabilities and patterns of hinge-point forces under different operating conditions are analyzed, and the perception characteristics of each hinge-point are clarified. Hinge-points A, A', B, and B', which are farther from the forging clamping position, generally lack the ability to perceive various aspects of the forging's force. In contrast, hinge-points O, O', C, and C' exhibit stronger perception capabilities. Moreover, the effectiveness of the x and y-axis components of the hinge-point forces varies under different operating conditions. The specific characteristics are as follows:
 - a. Changes in the rotational posture of the clamping mechanism can be perceived using the x-axis force components of the upper hinge-points C and C'.
 - b. Variations in the forging diameter can be detected using the y-axis force components of hinge-points O and O'.
 - c. The axial component of the forging force can be perceived using the x-axis force components of the lower hinge-points O' and C' under typical clamping states.
 - d. The vertical component of the forging force can be perceived using the x-axis force components of the upper hinge-points O and C under typical clamping states.

In the future, based on the force perception capabilities and characteristics of the hinge-points in the clamping mechanism, we will conduct research on active compliant control to reduce the impact and damage caused to the forging operation machine during the forging process, thereby improving the forging quality.

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Disclosure statement

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